

Zejun Wu

Contact Address: Imperial College London Silwood Park Campus, SL5 7PY

Education

- **University of Virginia (PhD)** 09/2024-now
 - Topic: Remote sensing and LiDAR on terrestrial ecosystem (forest)
- **Imperial College London – Ecological Application (Msc)** 10/2022-10/2023
 - Grades: *Distinction*
 - Core modules: Data Science; Field Ecology; Agroecology; Global Challenges; Master Project etc.
- **University of Liverpool**
Xi'an Jiaotong-Liverpool University – Environmental Science (BSc) 09/2018-07/2022
 - Grades: *First-class Honors*
 - Main courses: Environmental Chemistry, Spatial Modeling with GIS & RS, Soil Environment, etc.

Honors & Awards

- **Awards:**
 - 2023 Best Position Paper at the 8th Climate Change Conference of Parties (1/35)
 - 2021/2022 University Academic Excellent Award (1/24)
 - 2021 Summer Undergraduate Research Fellow
- **Publication:**
 - **Wu., et al.** Improving leaf spring phenology modeling for temperate tree species: An integration of the Farquhar-Medlyn photosynthesis model with the optimality-based approach. *Global Change Biology* (Co-first author, under review)
 - **Wu., et al.** Effects of selenite on the responses of lettuce (*Lactuca sativa* L.) to polystyrene nano-plastic stress. *Ecotoxicology and Environmental Safety*, 2023.
(First author; IF=7.13; Q1; <https://doi.org/10.1016/j.ecoenv.2023.115138>)
 - **Wu., et al.** Does the antagonistic effect of Se on Cd depend on their molar ratio in crops? A meta-analysis. *Environmental and Experimental Botany* (First Author; under review).
- **Invention Patent (in China):**
 - A device and method for toxicological study of plant roots (202111299459.2; First Inventor)

Research Experiences

Terrestrial ecosystem: vegetation & carbon cycle - Prof. Colin Prentice

- **Five-Months Master Research Project** 12.2022-11.2023
Comparison of estimation and seasonal cycle of Gross Primary Production (GPP) in Europe
 - Predicted GPP values by the P model using FPAR or proxied by SIF, and compared with in-situ GPP;
 - Detected the start and end time of season from GPP values by first-derivative, percentile-threshold, or logistical function methods;
 - Compared the consistency and difference of phenological indicators and explored the underlying factors.

Soil-plant system: plant ecotoxicology & trace-nutrient – Prof. Linxi Yuan

- **Final Year Project (published)** 09.2021-06.2022
*Effect of selenite (Se) on the response by lettuce (*Lactuca sativa* L.) under nano-plastics (NPs) stress*

- Designed the whole project; completed the experiment, analysis, and final reported individually;
 - For experiment: cultured lettuces for Se-NPs exposure; measured indicators (e.g., root morphology, Se concentration by ICP-MS, chlorophyll content by spectrophotometer, and enzyme activities);
 - For analysis: ANOVA for group differences; and PCA and Regression Path Analysis for mechanisms.
- **Independent Meta-analysis Project (under review)** 12.2021-06.2022
Does the antagonistic effect of Se on Cd depend on their molar ratio in crops? A meta-analysis
 - Searched and screened literatures, and extracted data from 46 papers from 1995 to 2022;
 - Conducted a meta-analysis by traditional learning model and wrote research paper.
 - **Leader for Summer Undergraduate Research Fellowships (patent)** 06.2021-09.2021
A smart device "Toxicology Testing Pen" to evaluate maize root response to Se/Cd interaction
 - Designed toxicological device and completed patent application report, independently;
 - Led team to complete the lab experiments, e.g., making root transverse sections, and data analysis.

Working Experiences

- **The University of Hong Kong** 10.2023-07.2024
Research assistant in department of Biological Sciences (Dr. Jin Wu)
 - Developing a new process-based model to test the eco-evolutionary optimality hypotheses on leaf-out;
 - Exploring biotic factors in regulating spatial variability of phenology across forest communities;
- **Ministry of Ecology and Environment of China** 02.2022-02.2023
Internship in soil environmental protection (part-time)
 - Surveyed the polluted soil sites and evaluated the current condition of soil pollution;
 - Simulated and predicted the scope and degree of pollution in the future;
 - Wrote the regional soil investigation and risk assessment report, especially in Hunan Province.
- **Jinan Water Group Co, Ltd** 01.2021-02.2021
Internship in Water Quality Management
 - Mastered process and technology of drinking water production; guaranteed safety of water supply;
 - Completed daily water quality test and report; revised water quality management.

Extracurricular Activities

- 2023 Bugs, Birds and Beasts Education – volunteer
- 2021 Young Scientist Forum of Earth Science – Participant
- 2019-2020 Education Support of summer social practice – Leader
- 2019 World Artificial Intelligence Conference – Excellent Volunteer
- 2018-2019 Hosting and broadcasting group of art association – Deputy Head
- 2018-2019 XJTLU Collegiate Debate Champion

Other Skills

- **Language:** English(fluent), Mandarin(native)
- **IT skills:** R, python, ArcGIS, ENVI, Microsoft Office suite, image J, and SPSS.
- **Lab skills:** master ICP-MS, spectrophotometer, portable photosynthetic machine, root scanning analyzer...